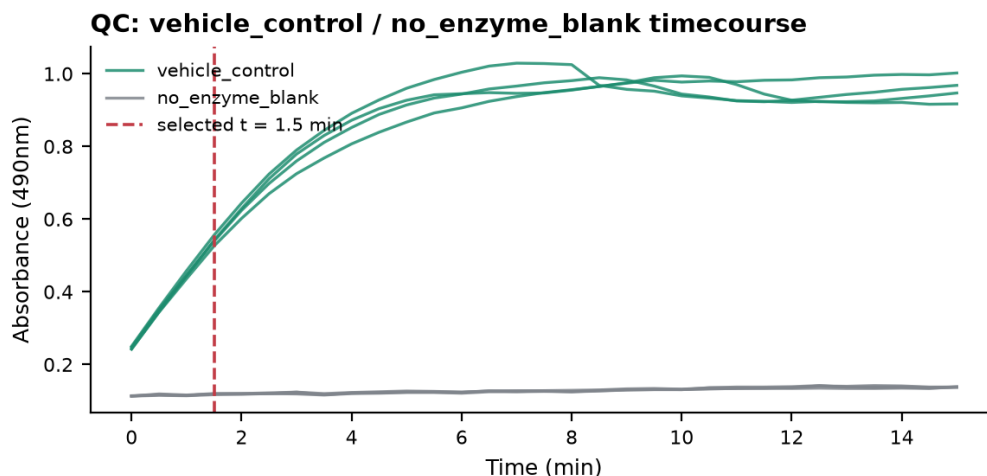


Experiment 2 — Results & IC50 Fits

100 μ L purified-enzyme dose-response, 5 compounds $\times$ 4 doses $\times$ n=2. TEM-1 β -lactamase / nitrocefin kinetic assay.

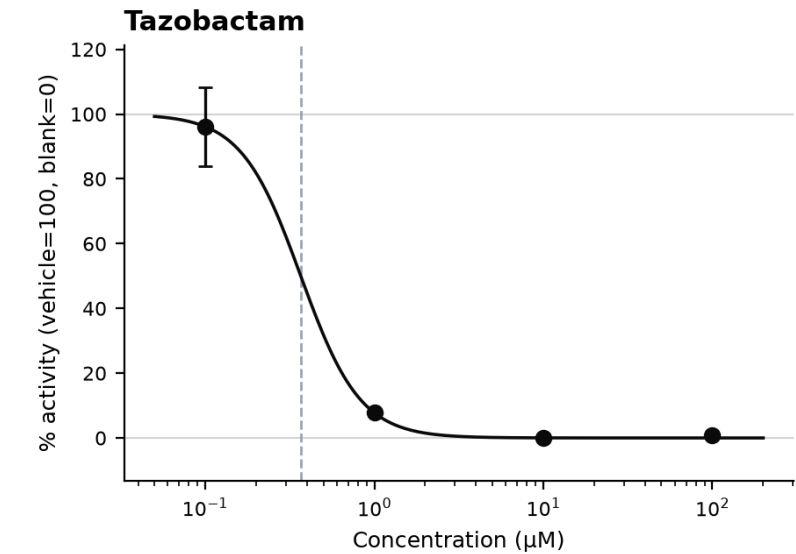
Plate QC

Single-timepoint (endpoint) read at t = 1.5 min, not a full-trace kinetic regression — chosen as the latest point still inside vehicle_control's own initial linear phase (see plot below). Reading later risks the fully-active well saturating the 200 μ M nitrocefin before a partially-inhibited well catches up, which would compress the apparent difference between them and understate potency.



vehicle_control mean signal: 0.2953 Δ Abs (n=4, defines 100% activity). no_enzyme_blank mean signal: 0.0055 Δ Abs (n=4, defines 0% activity floor). Assay window: 0.2898 Δ Abs.

Tazobactam — Control (lit. Ki 17nM)



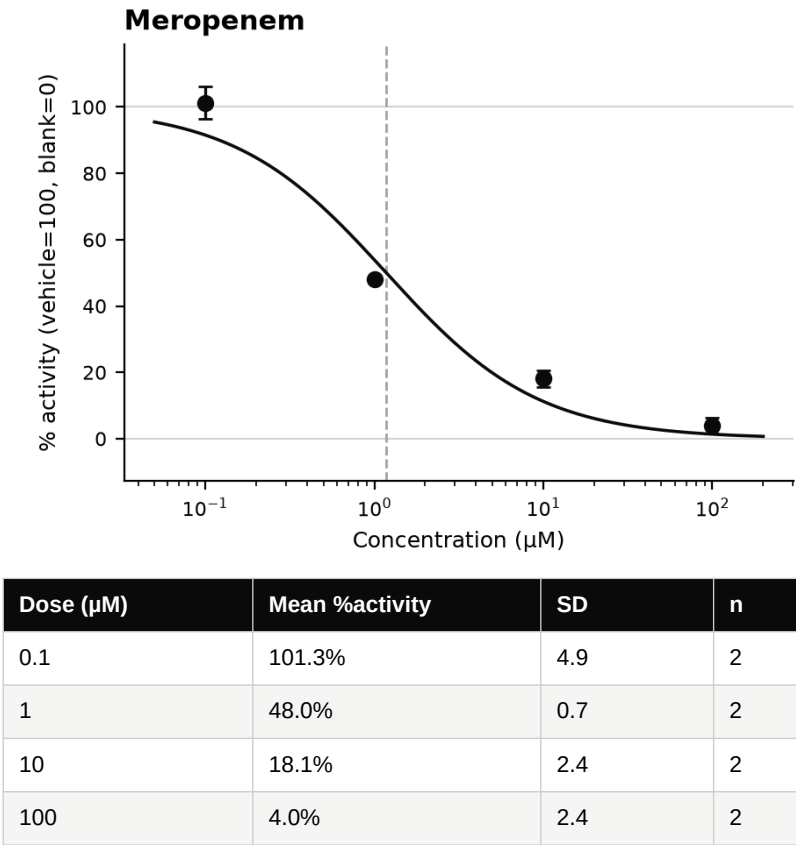
Dose (μM)	Mean %activity	SD	n
0.1	96.1%	12.2	2
1	7.8%	1.0	2
10	0.0%	0.2	2
100	0.9%	0.5	2

Anchored 2-parameter fit (top/bottom pinned to vehicle/blank): IC50 = 0.367 μM (95% CI 0.223–0.603 μM (t-distribution, 6 df)), Hill n = 2.47, R² = 0.989.

Nonparametric bootstrap (resample-with-replacement, n=977 successful refits of 1000): 90% CI 0.269–0.621 μM. With only n=2 replicates per dose, treat this as a rough first-pass estimate, not a validated potency value — a prior update, not a conclusion.

Free 4-parameter cross-check: top=96.1%, bottom=0.4%, IC50=0.608 μM, n=5.00. Only 8 data points for 4 free parameters — thin fit, shown as a cross-check on the anchored pinning above, not as the primary estimate.

Meropenem — Primary

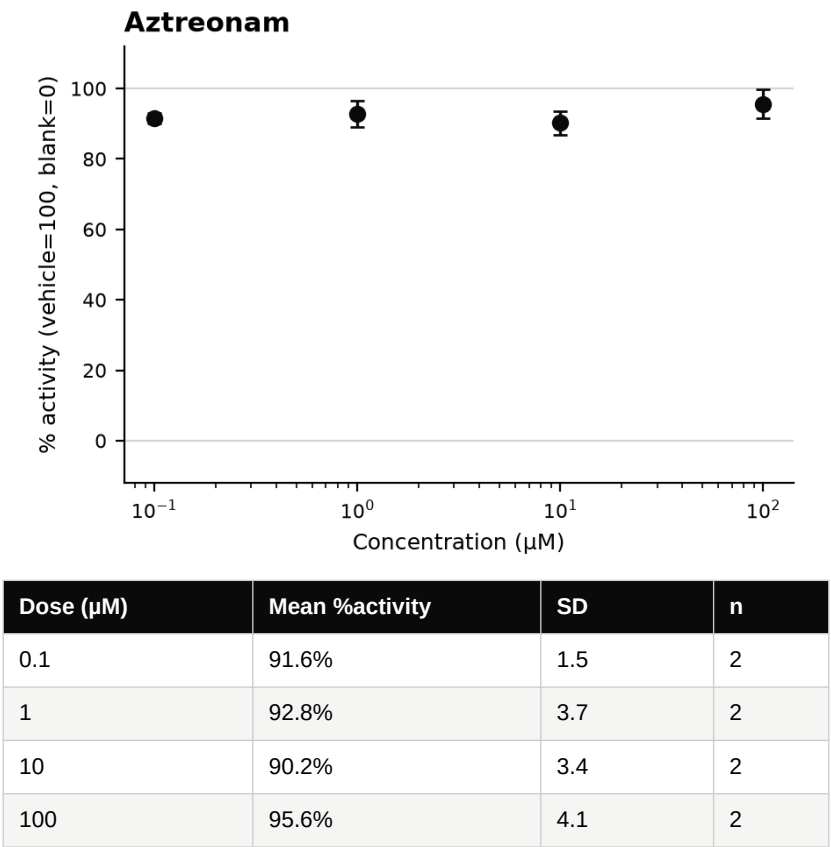


Anchored 2-parameter fit (top/bottom pinned to vehicle/blank): $\text{IC}_{50} = 1.18 \mu\text{M}$ (95% CI 0.694–2 μM (t-distribution, 6 df)), Hill $n = 0.97$, $R^2 = 0.964$.

Nonparametric bootstrap (resample-with-replacement, $n=970$ successful refits of 1000): 90% CI 0.896–3.42 μM . With only $n=2$ replicates per dose, treat this as a rough first-pass estimate, not a validated potency value — a prior update, not a conclusion.

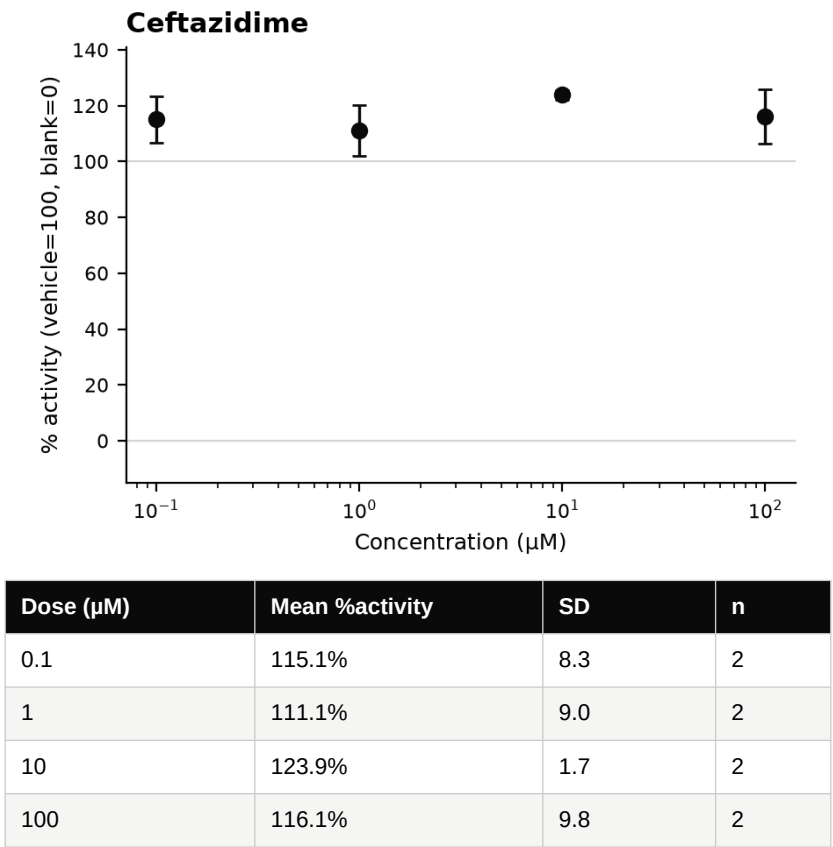
*Free 4-parameter cross-check: top=140.0%, bottom=2.3%, $\text{IC}_{50}=0.382 \mu\text{M}$, $n=0.68$. Only 8 data points for 4 free parameters — thin fit, shown as a cross-check on the anchored pinning above, not as the primary estimate. **Disagrees with the anchored fit by 0.3×** — the anchored pinning (top=100/bottom=0) may not be justified by this compound's actual data, especially if it's flagged above/below range.*

Aztreonam — Primary



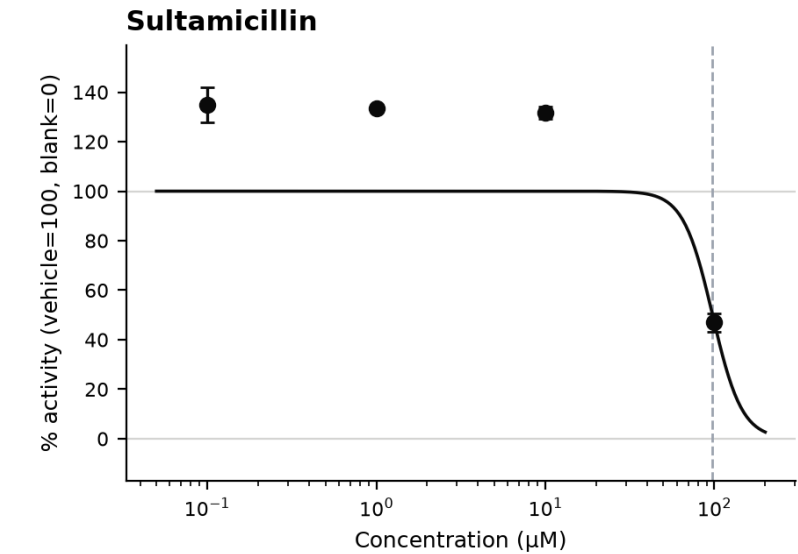
IC50: not determinable within the tested range (0.1–100µM). Mean activity at the top dose stayed above 70% — no meaningful inhibition observed in this screen.

Ceftazidime — Primary



IC50: not determinable within the tested range (0.1–100µM). Mean activity at the top dose stayed above 70% — no meaningful inhibition observed in this screen.

Sultamicillin — Expected-inert check



Dose (µM)	Mean %activity	SD	n
0.1	134.9%	7.1	2
1	133.4%	0.5	2
10	131.7%	2.4	2
100	46.9%	3.7	2

Anchored 2-parameter fit (top/bottom pinned to vehicle/blank): IC50 = 97.6 µM (95% CI unresolved (fit poorly constrained by the tested doses)), Hill n = 5.00, R² = 0.402.

Nonparametric bootstrap (resample-with-replacement, n=968 successful refits of 1000): 90% CI 95.6–934 µM. With only n=2 replicates per dose, treat this as a rough first-pass estimate, not a validated potency value — a prior update, not a conclusion.

Free 4-parameter cross-check: top=134.2%, bottom=-30.0%, IC50=93.4 µM, n=1.86. Only 8 data points for 4 free parameters — thin fit, shown as a cross-check on the anchored pinning above, not as the primary estimate.